

MATH – 241
Engineering Mathematics I
Problem Set – 9

Question 1. A necessary and sufficient condition for positive definiteness of a quadratic form $f(\underline{x}) = \underline{x}^T A \underline{x}$ with symmetric matrix A is that all the principal minors are positive, that is;

$$a_{11} > 0, \begin{vmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{vmatrix} > 0, \begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix} > 0, \dots, \det(A) > 0.$$

- a) Show that $4x_1^2 + 12x_1x_2 + 13x_2^2 = 16$ is positive definite.
- b) Show that $-11x_1^2 + 84x_1x_2 + 24x_2^2 = 156$ is indefinite. (Takes both positive and negative values.)

Question 2. Derive the general solution of

$$y_{n+3} + y_{n+2} - 4y_{n+1} + 4y_n = 0 \text{ with } y_0 = 3, y_1 = 5, y_2 = 9.$$

Question 3. Show that if $y_n^{(1)}$ and $y_n^{(2)}$ are solutions of the 2nd order linear homogeneous equation $a_0y_{n+2} + a_1y_{n+1} + a_2y_n = 0$, and Y_n is any particular solution of the non-homogeneous equation $a_0y_{n+2} + a_1y_{n+1} + a_2y_n = f_n$, then $y_n = C_1y_n^{(1)} + C_2y_n^{(2)} + Y_n$ is a solution of the non-homogeneous equation.

Question 4. Using question 3, solve;

$$y_{n+4} - 7y_{n+2} + 12y_n = n + 6 .$$